

MINECOFIN Rwanda fiscal projection and simulation model

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Basic properties of the model

Equations



General properties

Semi-structural "FPAS-type" gap model

Five main blocks: demand, prices, monetary policy, exchange rate, fiscal policy

Four price indicators: core, food, energy, headline CPI

Taylor rule with FX targeting

Hybrid UIP with FX targeting

Deficit targeting with automatic stabilizer effect

Debt financing rule

Fiscal impulse is determined by the primary deficit

Basic properties I: domestic block

Demand

Inflation There are two domestic prices, core and headline CPI. The core/CPI relative price is decomposed into an (exogenous) trend, and a gap.

RER and RIR The real exchange rate and the real interest rate are defined in terms of core prices.

Trends Potential growth, trend RER, the premium and US RIR trend are given by AR(1) processes. The Jordanian RIR trend is given by the real UIP.

Basic properties of the model

Equations



IS curve

$$\tilde{y}_t = \alpha_1 \tilde{y}_{t-1} + \alpha_2 \tilde{y}_{t+1|t} - \alpha_3 rmci_t + \alpha_4 fisc_t + \alpha_5 y\tilde{w}_t + \varepsilon_t^y \quad (1)$$

- $\tilde{y}_t$ is the output gap
- $rmci_t$ is the real monetary condition index, given as

$$rmci_t = \alpha_6 \tilde{r}4_t + (1 - \alpha_6) r\tilde{e}er_t \quad (2)$$

- $\tilde{r}4_t$ is four-period-ahead forward looking average real interest rate gap

$$\tilde{r}4_t = (\tilde{r}_t + \tilde{r}_{t+1|t} + \tilde{r}_{t+2|t} + \tilde{r}_{t+3|t})/4 \quad (3)$$

- $r\tilde{e}er_t$ is the real effective exchange rate gap
- $y\tilde{w}_t$ is effective world output gap

Phillips curve

$$\pi_t^{\text{core}} = \beta_1 \pi_{t-1}^{\text{core}} + (1 - \beta_1) \pi_{t+1|t}^{\text{core}} + \beta_2 rmc_t + \beta_3 (r\tilde{e}er_t - r\tilde{e}er_{t-1}) + \varepsilon_t^{\text{core}} \quad (4)$$

- π_t^{core} is core inflation
- rmc_t is the real marginal cost, given as

$$rmc_t = \beta_4 \tilde{y}_t + (1 - \beta_4) r\tilde{e}er_t \quad (5)$$

- $r\tilde{e}er_t$ is effective real exchange rate gap

UIP

$$s_t = \eta_1 (s_t^{\text{exp}} - (i_t - i_t^* - \text{prem}_t) / 4) + (1 - \eta_1) (s_{t-1} + \Delta s_t^{\text{tar}} / 4) + \varepsilon_t^s \quad (6)$$

- s_t is (log of) the USDRWF exchange rate (depreciation is up)
- i_t^* is the Fed Funds Rate
- Δs_t^{tar} is the target nominal depreciation
- s_t^{exp} is the expected exchange rate, given as

$$s_t^{\text{exp}} = \eta_2 (s_{t-1} + 2 (\Delta \bar{z}_t + \overline{\pi 4}_t^{\text{core}} - \pi^*) / 4) + (1 - \eta_2) s_{t+1|t} \quad (7)$$

- $\Delta \bar{z}_t$ is the trend real depreciation
- π^* is the US inflation target
- $\overline{\pi 4}_t^{\text{core}}$ is the (implied) one-year-ahead domestic CPI inflation target (y-o-y)

Monetary policy

$$i_t = \gamma_1 i_{t-1} + (1 - \gamma_1) \left(\bar{i}_t + \gamma_2 \left(\pi_{4_{t+4|t}} - \overline{\pi}_{4_t} \right) + \gamma_3 \tilde{y}_t + \gamma_4 (\Delta s_t - \Delta s_t^{\text{tar}}) \right) + \varepsilon_t^i \quad (\text{IFB})$$

- Taylor rule with FX targeting
- The target depreciation rate is consistent with PPP in the long run

$$\Delta s_t^{\text{tar}} = \eta_4 \Delta s_{t-1}^{\text{tar}} + (1 - \eta_4) \left(\Delta \bar{z}_t + \overline{\pi}_{4_t}^{\text{core}} - \pi^* \right) \quad (8)$$

- $\bar{i}_t$ is the sum of the inflation target and the trend real interest rate, which is derived from the UIP in real variables

$$\bar{i}_t = \bar{r}_t + \overline{\pi}_{4_t}^{\text{core}} \quad (9)$$

$$\bar{r}_t = \bar{r}_t^* + \text{prem}_t - \Delta \bar{z}_{t+1|t} \quad (10)$$

Food and energy prices

$$\tilde{r}p_t^{\text{food}} = \phi_{11} \tilde{r}p_{t-1}^{\text{food}} + \phi_{12} \left(\tilde{z}_t + \tilde{r}p_t^{\text{food},*} \right) + \varepsilon_t^{rp^{\text{food}}} \quad (11)$$

$$\tilde{r}p_t^{\text{ener}} = \phi_{21} \tilde{r}p_{t-1}^{\text{ener}} + \phi_{22} \left(\tilde{z}_t + \tilde{r}p_t^{\text{ener},*} \right) + \varepsilon_t^{rp^{\text{ener}}} \quad (12)$$

- Relative food/energy prices are modelled
- Depending on the world relative food and energy prices
- Note that we have from these definitions that

$$\begin{aligned} z_t + rp_t^{\text{food},*} &= s_t + p_t^* - p_t^{\text{core}} + p^{\text{food},*} - p_t^* \\ &= s_t + p^{\text{food},*} - p_t^{\text{core}} \\ &= s_t + p^{\text{food},*} - p_t^{\text{food}} + p_t^{\text{food}} - p_t^{\text{core}} \\ &= z_t^{\text{food}} + rp_t^{\text{food}} \end{aligned} \quad (13)$$

- This formulation avoids the complication with relative prices in the Phillips curves

Fiscal policy

- Fiscal policy is represented by the following variables
 - overall budget deficit (in percent of GDP)
 - stock of debt (in percent of GDP)
 - interest rate on debt
- All three variables are defined for both local and foreign currency
- There's a long term deficit target
- The automatic stabilizer captures the cyclical nature of the budget (mostly attributable to revenues)
- Fiscal impulse is defined in terms of the primary deficit

Debt and interest payments

- Debt evolves according to the familiar identities

$$debt_t^{lcy} = debt_{t-1}^{lcy} \frac{1}{\left(1 + \frac{\Delta\pi_t}{400}\right) \left(1 + \frac{\Delta y_t}{400}\right)} + def_t^{lcy} \quad (14)$$

$$debt_t^{fcy} = debt_{t-1}^{fcy} \frac{1 + \frac{\Delta S_t}{400}}{\left(1 + \frac{\Delta\pi_t}{400}\right) \left(1 + \frac{\Delta y_t}{400}\right)} + def_t^{fcy} \quad (15)$$

- Interest payments are then given as

$$intp_t^{lcy} = \frac{i_{t-1}^{debt, lcy}}{400} debt_{t-1}^{lcy} \frac{1}{\left(1 + \frac{\Delta\pi_t}{400}\right) \left(1 + \frac{\Delta y_t}{400}\right)} \quad (16)$$

$$intp_t^{fcy} = \frac{i_{t-1}^{debt, fcy}}{400} debt_t^{fcy} \frac{1 + \frac{\Delta S_t}{400}}{\left(1 + \frac{\Delta\pi_t}{400}\right) \left(1 + \frac{\Delta y_t}{400}\right)} \quad (17)$$

Deficit decomposition

- Overall deficit is written as

$$def_t = def_t^{\text{struct}} + def_t^{\text{cyc}} + def_t^{\text{discr}} \quad (18)$$

- The cyclical component (a.k.a. automatic stabilizer) is tied to the output gap

$$def_t^{\text{cyc}} = \delta_1 \tilde{y}_t \quad (19)$$

- The stability of the deficit is ensured by the equation for the structural component

$$def_t^{\text{struct}} = \delta_4 def_{t-1}^{\text{struct}} + (1 - \delta_4) + \overline{def}^{\text{struct}} + \varepsilon_t^{\text{def}^{\text{struct}}} \quad (20)$$

- The discretionary component represents temporary spending decisions of the government

$$def_t^{\text{discr}} = \delta_2 def_{t-1}^{\text{discr}} + \varepsilon_t^{\text{def}^{\text{discr}}} \quad (21)$$

Fiscal impulse

- We first define the *adjusted structural deficit* as

$$def_t^{\text{struct, adj}} = def_t^{\text{struct}} - intp_t^{\text{fcy}} - intp_t^{\text{lcy}} \quad (22)$$

- Fiscal impulse is then given as

$$fisc_t = \delta_3 def_t^{\text{discr}} + (1 - \delta_3) \Delta def_t^{\text{struct, adj}} \quad (23)$$

- This definition is consistent with the interpretation that interest payments are part of the structural expenditures
- The formula ensures that an increase in the deficit due to higher interest rates is not misinterpreted as fiscal stimulus

Expenditure/revenue decompositions

- The above decomposition can also be interpreted in terms of expenditures and revenues
- Expenditures are either structural or discretionary
 - Government wages, subsidies, etc.
 - Ad hoc transfers, disaster relief, election spending, etc.

$$exp_t = exp_t^{\text{struct}} + exp_t^{\text{discr}} \quad (24)$$

- Revenues are either structural or cyclical
 - Structural revenues depend on potential GDP and tax rates
 - Cyclical revenues respond to the output gap

$$rev_t = rev_t^{\text{struct}} + rev_t^{\text{cyc}} \quad (25)$$

Matching the components

- The components in the two decompositions can be paired as follows

$$def_t^{\text{struct}} = exp_t^{\text{struct}} - rev_t^{\text{struct}} \quad (26)$$

$$def_t^{\text{cyc}} = -rev_t^{\text{cyc}} \quad (27)$$

$$def_t^{\text{discr}} = exp_t^{\text{discr}} \quad (28)$$

- We can also write, using the notation of the ToR

$$exp_t^{\text{struct}} = G_t^{\text{struct}} + IG_t^{\text{struct}} \quad (29)$$

$$exp_t^{\text{discr}} = G_t^{\text{discr}} + IG_t^{\text{discr}} \quad (30)$$

$$rev_t^{\text{struct}} = T_t^{\text{struct}} + NFG_t \quad (31)$$

$$rev_t^{\text{cyc}} = T_t^{\text{cyc}} \quad (32)$$

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